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RESEARCH ARTICLE

Human Pose Estimation and Event Recognition via Feature Extraction and Neuro-Fuzzy Classifier

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ABSTRACT This paper introduces an advanced framework for Human Pose Estimation (HPE) and Semantic Event Classification (SEC), addressing the growing demand for sophisticated human skeleton modeling, context-aware feature extraction, and machine learning techniques for accurate event recognition in real-world scenarios. HPE, a fundamental task in applications such as sports analytics and surveillance systems, involves predicting human joint locations from visual data. Despite substantial progress driven by deep learning advancements, particularly in handling occlusions and crowded environments, a comprehensive integration of these innovations remains limited in current literature. To bridge this gap, we propose a novel HPE and SEC system featuring a robust six-stage pipeline. The workflow begins with preprocessing steps, including video-to-image sequence conversion, floor removal to reduce noise, grayscale transformation, and human silhouette extraction via binary masks. Human detection is performed using the GrabCut algorithm. Full-body feature extraction incorporates 1-D Fourier Transform, 2-D energy-based motion analysis, 0–180° intensity mapping via the Radon Transform algorithm, and movable body part detection using Fourier-Based Correlation. Key point features are characterized through the degree of freedom, human landmark detection via the HSV algorithm, and angular point analysis using the media pipe algorithm. Features are then optimized using the Ray Optimizer and classified through a Neuro-Fuzzy classifier. The proposed system achieves classification accuracies of 94.82%, 92.33%, and 93.63% on the UCF Sports Actions, Human Action Recognition (HAR), and MPII Human Pose datasets, respectively, outperforming state-of-the-art methods and demonstrating its effectiveness in tackling real-world challenges.

INDEX TERMS Pose estimation, GrabCut, semantic event classification, decision making, feature extraction, neuro-fuzzy classifier, machine learning.

I. INTRODUCTION

In the digital age, the proliferation of multimedia data has transformed how we analyze and interpret human-centric visual content such as images and videos. The combination of data visualization, improved image processing, and machine learning has created new opportunities for designing intelligent systems that can monitor and recognize human behaviors in various settings [1]. These systems have various applications, including surveillance, healthcare, education, crime prevention, and smart infrastructure. HAR, an important topic in this science, is concerned with recognizing and

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categorizing human posture and motions from real-time films or static photographs [2]. This skill is critical for tackling today's concerns, such as the growing need for automated solutions in patient monitoring, violence detection, sports performance assessment, and security surveillance [3]. Conventional approaches relying on human operators to monitor video streams are inherently limited by cost, time, and error rates, highlighting the demand for automated HAR systems capable of real-time, accurate, and cost-effective analysis [4].

This study describes a unique approach to HPE and SEC that aims to improve HAR systems by employing cutting-edge approaches in feature extraction, optimization, and classification. The proposed system combines sophisticated preprocessing approaches for visual input, accurate human posture modeling with novel feature extraction methods, and a robust classification pipeline based on a Neuro-Fuzzy classifier. Compared to benchmark datasets, the framework outperforms at detecting human behaviors in a variety of real-world contexts [6]. By automating the analysis of human behavior and identifying routine or suspicious activities, the proposed approach not only advances HAR research but also contributes significantly to applications such as smart homes, healthcare monitoring, and security. This work addresses critical gaps in existing literature, particularly the lack of comprehensive integration of modern deep learning and machine learning techniques for precise and scalable human activity analysis. The major concerns of our work can be highlighted as follows:

- We propose an automated framework for human pose estimation and event recognition that incorporates advanced preprocessing techniques, including floor removal, grayscale transformation, and binary masking, ensuring robust and precise human silhouette extraction.
- We introduce a novel multi-level feature extraction method that leverages motion analysis, intensity mapping, and movable body part detection, alongside key point characterization using media pipe and HSV algorithms, for comprehensive and distinctive feature representation.
- We optimize feature selection using the Ray Optimizer and employ a Neuro-Fuzzy classifier to ensure robust and accurate classification of human activities and events.
- We demonstrate the effectiveness of the proposed system on publicly available datasets, showing its superior performance compared to state-of-the-art methods across diverse scenarios.

The structure of the paper is organized as follows: Section II provides a comprehensive review and critical analysis of related literature. Section III elaborates on the proposed methodology. Section IV introduces the dataset employed in the study and validates the system's reliability through extensive trials. Section V discusses the limitations encountered in the approach. Finally, Section VI summarizes the findings and proposes directions for future research. The complete workflow of the proposed method is illustrated in Fig. 1.

II. RELATED WORK

Significant advancements in computer vision and machine learning have enabled researchers to develop robust systems for HPE and HAR. These efforts primarily focus on utilizing vision-based approaches, leveraging spatiotemporal features from video data to understand human movements and behaviors. Recent innovations include multi-level feature extraction techniques, optimization algorithms, and advanced classifiers, aiming to enhance accuracy and reliability in dynamic and complex environments. Such approaches have become pivotal in applications like surveillance, healthcare, and sports analytics, motivating the exploration of more efficient and scalable methodologies.

A. TRADITIONAL APPROACHES TO HUMAN POSE ESTIMATION

Zhen and Shao [7] investigated several local computing approaches for HAR, such as STIPs, bag-of-words, sparse coding, Improved Fisher's Kernel (IFK), and VLAD. They used a naïve Bayes nearest neighbor algorithm for classification. While the research lacks a formal architectural framework for human action recognition, it does provide a comparative analysis that identifies IFK as the most effective for local computations. In contrast, the suggested system provides an end-to-end design that produces greater outcomes. Wang [8] suggested a Hidden Markov Model (HMM)-based technique for HAR that effectively models sequential data. While the technique works well for distinguishing basic activities, its accuracy suffers when applied to complicated or overlapping behaviors owing to its poor adaptation to high-dimensional data. Einfalt et al. [9] designed a unified method for recognizing sports events involving an athlete in motion, using task classification, extraction of chronological 2D posture features, and a convolutional sequence network. This approach enabled precise sports event recognition. However, the method is limited by its reliance on specific motion patterns, which may not generalize well to a wide range of sports or activities, and its performance could be affected by factors such as occlusion or rapid, complex movements.

1) MACHINE LEARNING-BASED APPROACHES

Charith et al. [10] conducted a research to compare the performance of several posture classification models trained on data from a human position estimation framework, with an emphasis on combining pose estimation with machine learning for yoga pose classification. While the research sheds information on model performance and the potential of this integration, it is constrained by its reliance on high-quality pose estimate outputs, which may worsen in the presence of occlusion, low-light circumstances, or complicated postures. Furthermore, the technique is strongly dependent on controlled datasets, which may not translate well to different real-world settings. Gado et al. [11] explored a multimodal

data fusion approach combining acceleration and physiological signals to enhanced HAR. Their method demonstrated improved activity classification in controlled settings but faced scalability issues in complex, real-world environments. Cheng et al. [12] proposed a novel HAR system integrating virtual reality (VR) data with traditional sensing methods to classify movement characteristics, particularly in phobia scenarios like acrophobia. Their framework uses VR environments to simulate realistic contexts, enhancing the precision of activity recognition. However, the system's reliance on VR restricts its adaptability for general-purpose HAR applications outside specific phobia-related scenarios.

B. DEEP LEARNING-BASED APPROACHES

Liu et al. [13] offer ConvCCPose, an efficient human pose estimation model based on ConvNeXt that strikes a compromise between accuracy and computing economy. The model employs a classification-based technique to independently localize horizontal and vertical coordinates, removing the requirement for post-processing and simplifying implementation. However, its enormous parameter count and significant latency provide issues for real-time industrial applications. Feng et al. [14] introduced a directed Long Short-Term Memory (LSTM) framework paired with a Convolutional Neural Network (CNN)-based model for extracting the temporal locations of deep features in composite movies. For person identification, the cutting-edge YOLOv3 model is utilized, while event recognition is done using the LSTM framework. However, the approach is limited by its reliance on high-quality input data, which can affect performance in scenarios with occlusions, complex backgrounds, or low-resolution videos. Shao et al. [2] present a densely connected recurrent network for event-based human pose estimation, addressing challenges unique to event cameras, which capture only dynamic pixel changes. The network enhances pose estimation accuracy and stability by modeling sequential and non-sequential geometric consistency across time steps. However, event cameras' focus on moving parts often neglects static body features, and their novel paradigm limits the available information, capturing only motion-driven signals. Despite these limitations, the approach effectively leverages temporal and geometric patterns for robust pose estimation. Jiang et al. [15] focus on 2D human pose estimation calibration and keypoint visibility classification, improving accuracy and aligning confidence scores with keypoint quality. However, the method does not address event classification and has limitations, as confidence scores can be unreliable in occlusion scenarios, and high confidence may not always indicate accurate keypoint estimation.

C. HYBRID APPROACHES

A lightweight method for event recognition that emphasizes optical flow, optimal allocation, and a histogram of extracted optical flow was created by Li et al. [16]. They accomplished successful event recognition by utilizing the Reduced and Compressed Coefficient Dictionary Learning (LRCCDL) methodology, body joint restoration, and a typical optimization process. The method's limitations include its dependence on particular attributes, which might not translate well to complicated or varied event scenarios, and its potential to perform worse when noisy or low-quality data is present. Using a marker-based methodology, Jalal et al. [17] created a reliable system in which several body markers are affixed to the human body at strategic locations. The system successfully detects human behavior by utilizing a multi-actor platform and extracts synthetic settings. The technique is constrained, though, by its dependence on physical markers, which could be uncomfortable or impede natural movement. Additionally, its efficacy might be diminished in dynamic or uncontrolled settings where indicators are difficult to see or could move. For gesture-based activity recognition, Zheng et al. [18] presented a hybrid few-shot learning technique that makes use of biosensors. The model's capacity to manage tiny datasets efficiently is enhanced by the combination of sophisticated machine-learning techniques and conventional classifiers. However, its dependency on specialized biosensor hardware limits its scalability and broader application in diverse real-world environments. Yu et al. [19] present a heuristic framework for detecting events in soccer competition videos by using a replay identification approach to extract key contextual features. This method aims to meet spectator demands and create replay story clips. However, the framework is limited by its dependence on replay segments, which may not capture all significant events, and its effectiveness may decrease in scenarios with inconsistent video quality or editing styles. Shao et al. [2] propose a densely connected recurrent architecture to tackle incomplete information in event cameras by modeling both sequential and non-sequential geometric consistency over time. This approach enables the recovery of full human body poses, ensuring stable and accurate pose estimation from event data. However, the method is limited by the sparse nature of event signals, which capture only moving body parts, often leading to incomplete or missing body parts during certain time intervals.

As opposed to the limitations noted in the previous research, our study presents an advanced six-stage framework for HPE and Semantic Event Classification (SEC) that overcomes problems such as occlusion, noisy data, and the need for specialist equipment. Unlike previous methods that struggle with incomplete pose detection or rely on organized datasets, our approach incorporates strong preprocessing techniques such as video-to-image conversion, GrabCut-based human detection, and binary masking for silhouette extraction, resulting in noise reduction and accurate region identification. We perform dynamic and context-aware feature extraction by combining the 1-D Fourier Transform, 2-D Energy-based motion analysis, and moveable part identification techniques. Moreover, our system employs the Ray Optimizer for feature refinement and a Neuro-Fuzzy classifier for precise event recognition, surpassing state-of-the-art methods with classification accuracies across the

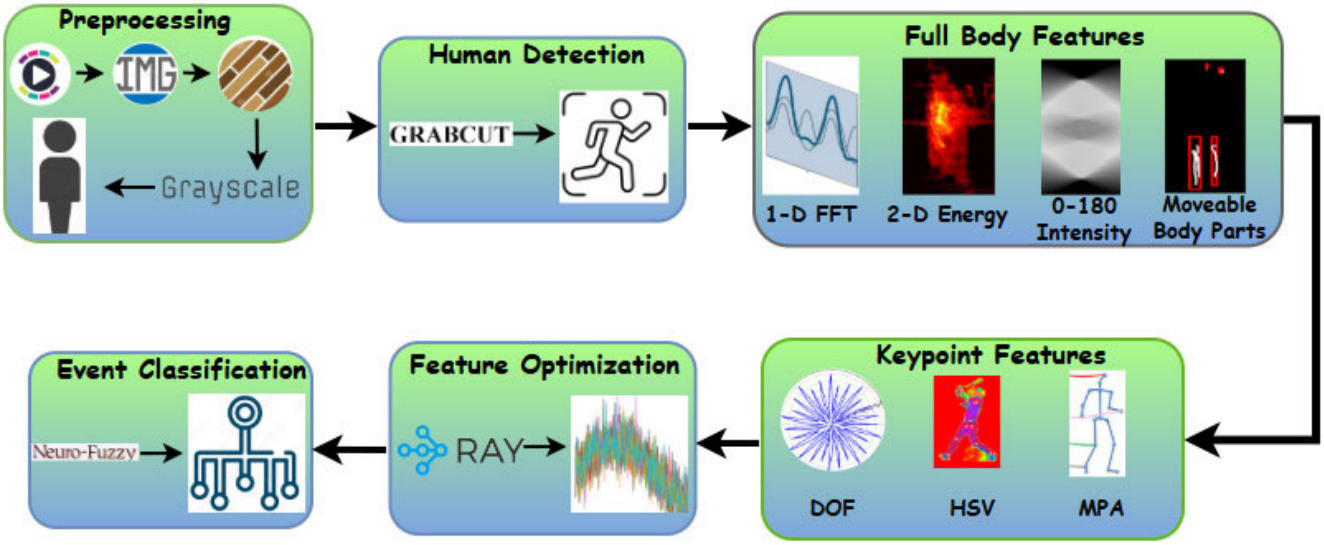

FIGURE 1. The architecture of the proposed human posture estimation and event classification.

UCF Sports Actions, HAR, and MPII datasets, respectively. This comprehensive and computationally efficient strategy addresses the shortcomings of earlier methods, delivering superior performance in diverse real-world scenarios.

III. SYSTEM METHODOLOGY

Our HPE and Semantic Event Classification (SEC) system leverages a robust six-stage pipeline to address real-world challenges. The process begins with preprocessing steps like video-to-image conversion, floor removal for noise reduction, and silhouette extraction to ensure clean data for further analysis. Human detection is performed using the GrabCut algorithm, which enables effective identification in cluttered or noisy environments. Feature extraction involves 1-D Fourier Transform, 2-D Energy-based motion analysis, intensity mapping through the Radon Transform algorithm, and detection of distinct movable body parts. Keypoint detection is enhanced through the HSV algorithm and angular point analysis, optimized with the media pipe algorithm. The features are then fine-tuned using the Ray Optimizer, and event classification is conducted using a Neuro-Fuzzy classifier to accurately categorize events based on the extracted features.

A. PREPROCESSING

The preprocessing stage of our proposed system involves several critical steps aimed at preparing the raw video data for effective analysis. First, the videos are converted into image sequences, enabling the system to process each frame independently and efficiently [20]. To reduce noise and enhance the quality of the images, a floor removal technique is applied, which blurs the background and removes unnecessary details, focusing the system's attention on the human subject. Following this, the images are converted to grayscale to simplify the data and reduce computational complexity, ensuring that the system can focus on structural features rather than color variations [21]. Finally, human silhouettes are extracted using advanced segmentation techniques, isolating the subject from the background and providing a clean representation of the person's pose and movements. This preprocessing pipeline ensures that the data entering subsequent stages is of high quality and relevance, setting the foundation for accurate pose estimation and event classification.

$$I_{frame}(t) = V\left(\frac{t}{f_s}\right) for\ t = 0, 1, 2, \ldots, N \tag{1}$$

where I_{frame} is the frame at time t, V represents the video stream, f_s is the sampling rate, *and* N is the total number of frames in the video.

$$I_{foreground}(x, y, t) = I(x, y, t) - BG(x, y, t) \tag{2}$$

where $I(x, y, t)$ is the intensity value of the image at pixel coordinates *(x,y)* at time t. $BG(x, y, t)$ is the background model at pixel coordinates *(x,y)* at time t, $I_{foreground}(x, y, t)$ is the foreground (human silhouette) obtained after background subtraction. The outcomes of the preprocessing steps are depicted in Fig. 2.

B. HUMAN DETECTION

Human detection is a critical step for accurate human pose estimation and event classification. The GrabCut algorithm, a graph-based image segmentation technique, is employed in our approach to isolate human figures from complex backgrounds [22]. GrabCut combines iterative graph cuts and pixel classification to segment images effectively, distinguishing foreground from background. It utilizes Gaussian Mixture Models (GMM) to model the color distributions of both foreground and background pixels, iteratively refining the segmentation. Through energy minimization techniques, GrabCut ensures optimal segmentation, making it especially

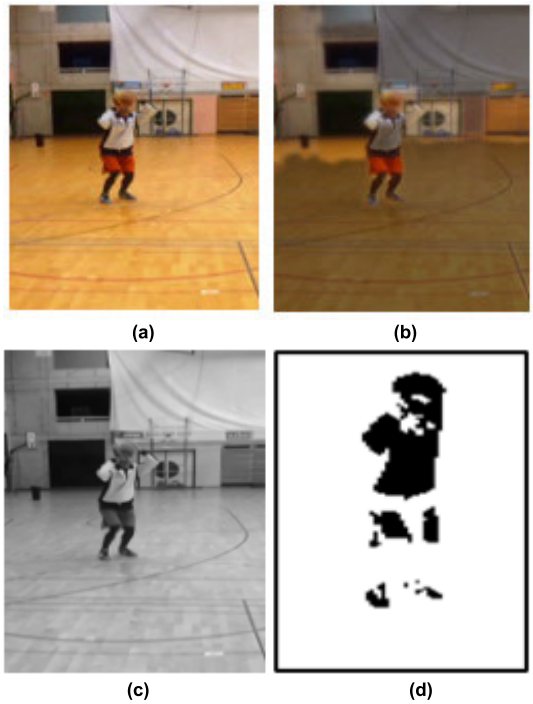

FIGURE 2. Preprocessing results (a) Original RGB image and (b) Floor removal image (c) Grayscale image and (d) Human silhouettes.

effective in noisy environments and scenes with dynamic lighting. This robustness allows for accurate segmentation, even in cluttered or occluded settings.

The algorithm iteratively improves segmentation by minimizing an energy function, ensuring that pixels classified as foreground or background maintain consistent color distributions. It partitions the image using graph cuts while considering local pixel intensity variations, based on a Markov Random Field (MRF) model, where pixels are represented as nodes and the probability of neighboring pixels sharing labels as edges [23]. The result is a precise segmentation of human figures, suitable for subsequent analysis such as feature extraction and pose estimation. GrabCut's ability to accurately segment humans, even in dynamic and challenging scenarios, greatly enhances detection performance in complex visual data.

$$E(F) = \sum_{i \in P} D_i\,(F_i) + \lambda \sum_{(i,j) \in N} V_{ij}(F_i, F_j) \tag{3}$$

where F is the segmentation of the image, $D_i\,(F_i)$ is the data term, representing the likelihood of pixel i belonging to either the foreground or the background. $V_{ij}(F_i, F_j)$ is the smoothness term, which encourages neighboring pixels to have similar labels, N represents the neighboring pixels, and λ is a weighting factor controlling the balance between the data and smoothness terms.

$$P\,(x_i \mid \theta_k) = \sum_{k=1}^{K} w_k . N(x_i;\, \mu_k;\, \Sigma_k) \tag{4}$$

where x_i is the color value of pixel I, $\theta_k = \{w_k;\, \mu_k;\, \Sigma_k\}$ represents the parameters of the k-th Gaussian component, $N(x_i;\, \mu_k;\, \Sigma_k)$ is the Gaussian distribution for pixel x_i, K is the number of Gaussian components used in the model.

FIGURE 3. Human detection via Grab-cut algorithm.

C. FULL BODY FEATURES

To enhance the accuracy and robustness of full-body feature extraction, our framework employs a combination of advanced analytical techniques like 1-D Fourier Transform, 2D Energy Heat Mapping, 0–180° Intensity Mapping, and Detection of Distinct Movable body parts. These methods collectively address the spatial, temporal, and angular attributes of human motion, ensuring a comprehensive representation of the body's dynamics [24]. Each technique is systematically integrated into the pipeline, leveraging specific algorithms to extract distinct features that support downstream tasks like pose estimation and event classification with heightened precision.

1) 1-D FOURIER TRANSFORM

The 1-D Fourier Transform, implemented using the Fast Fourier Transform (FFT) algorithm, plays a pivotal role in analyzing temporal patterns in human motion data. This technique decomposes time-domain signals into their frequency components, enabling the identification of key motion characteristics, such as repetitive or periodic behaviors [25]. By applying FFT to extracted features from human silhouettes, we transform spatial data into a frequency spectrum, highlighting dominant motion patterns that aid in downstream tasks like pose estimation and event classification [26]. This transformation is particularly valuable in isolating significant frequency components associated with distinct human activities, making it a robust tool for analyzing complex motion dynamics [27].

$$X\,(k) = \sum_{n=0}^{N-1} x\,(n).e^{-j\frac{2\pi}{N}kn},\, k = 0, 1, \ldots, N-1 \tag{5}$$

where $x\,(n)$ is the input time-domain signal, $X(k)$ is the frequency-domain representation, N shows the number of samples, $e^{-j\frac{2\pi}{N}kn}$ represents the Complex exponential kernel for frequency decomposition.

$$E\,(k) = |X\,(k)|^2 = Re\{X(k)\}^2 + Im\{X(k)\}^2 \tag{6}$$

where $E(k)$, represents the energy of a frequency component k in a signal. It is derived as the square of the magnitude $\{X(k)\}^2$, where $X(k)$ is the Fourier Transform output. The energy is computed by summing the squares of the real $Re\{X(k)\}^2$ and imaginary $Im\{X(k)\}^2$ parts of $X(k)$.

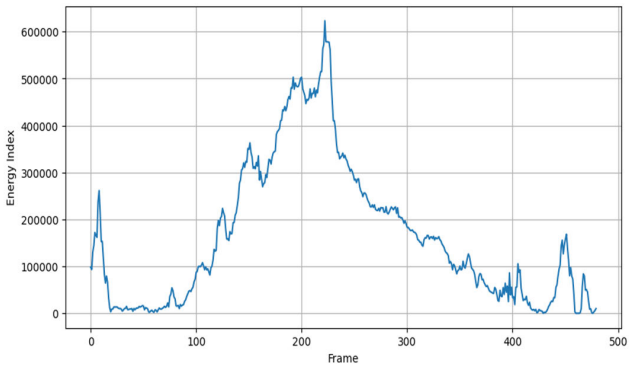

FIGURE 4. 1-D Fourier transform graph for the full body detected features.

2) 2-D ENERGY HEAT MAPPING

The 2D Energy Heat Graph provides a visual representation of motion intensity and spatial distribution, enhancing the analysis of human activity by highlighting areas with significant movement [28]. This technique is powered by Dense Optical Flow (Farneback) and MOG2 Background Subtraction, which work in tandem to extract and map motion patterns onto a heat graph. Dense Optical Flow computes the pixel-wise motion between consecutive frames, capturing both the magnitude and direction of movement. Meanwhile, MOG2 Background Subtraction isolates dynamic foreground elements by modeling and subtracting the static background, ensuring that the graph focuses exclusively on human motion [29]. By combining these algorithms, the 2-D Energy Heat Graph effectively highlights regions of interest, enabling the system to detect and quantify human activity with precision.

$$I_2(x+u, y+v) \approx I_1(x, y) + \nabla I \cdot \begin{bmatrix} u \\ v \end{bmatrix} \tag{7}$$

where (u,v) represents the motion vector at pixel (x,y), ∇I is the Image gradient representing intensity changes.

$$P(I_t) = \sum_{k=1}^{K} \omega_k . N(I_t; \mu_k; \sigma_k^2) \tag{8}$$

where $P(I_t)$ shows the probability of intensity I_t belonging to the foreground or background. ω_k shows the weight of the k-th Gaussian component. μ_k, σ_k^2 are the mean and variance of the k-th component.

$$E(x, y) = \sum_t \frac{M_t(x, y) . S_t(x, y)}{255} \tag{9}$$

where $M_t(x, y)$ shows the motion of the magnitude at pixel *(x,y)* in frame t. $S_t(x, y)$ represents the silhouette mask value (255 for foreground, 0 for background).

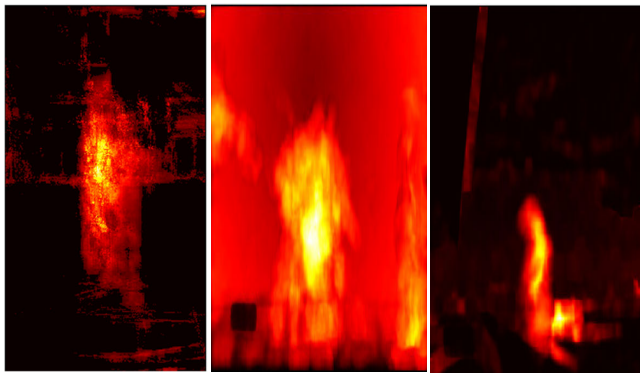

FIGURE 5. Result of 2D-Energy Heat features over the image.

3) 0–180° INTENSITY MAPPING

The Radon Transform technique, a potent mathematical tool for examining projections of intensity values across certain angles, is used in the 0–180° intensity mapping. Comprehensive motion analysis is made easier by this technique, which is especially useful for recording and displaying the directional distribution of intensity variations within the human body silhouette [30]. The technique maps the cumulative intensity of pixels along predetermined angle projections, ranging from 0° to 180°, by converting an image's spatial domain into its Radon domain. This makes it possible to identify the main movement directions and areas, which is essential for describing dynamic human activities [31].

The Radon Transform processes the image *I(x,y)* by integrating intensity values along straight lines at various angles θ. These projections encode the structure of the image, emphasizing linear and angular features. The transformed data, often referred to as a sinogram, provides a compact representation of the image's intensity distribution, which is further processed for feature extraction [32]. By leveraging the Radon Transform, our approach effectively captures nuanced intensity variations and directional patterns, enhancing the robustness of the feature set for downstream tasks like classification and event recognition.

$$R(\rho, \theta) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(x, y)\delta(\rho - x\cos\theta - y\sin\theta)dxdy \tag{10}$$

where $R(\rho, \theta)$ is the projection of the intensity function *I(x,y)* at angle θ, $\rho = x\cos\theta + y\sin\theta$ shows the perpendicular distance from the origin to the line of projection, δ is the Dirac delta function ensuring integration along the line defined by ρ and θ.

$$R(\rho, \theta) = \int_{-\infty}^{\infty} F(u, v)e^{j2\pi\rho(u\cos\theta + v\sin\theta)}dudv \tag{11}$$

where, *F(u,v)* is the Fourier Transform of *I(x,y)*, and the exponential term encodes the angular orientation in the frequency domain.

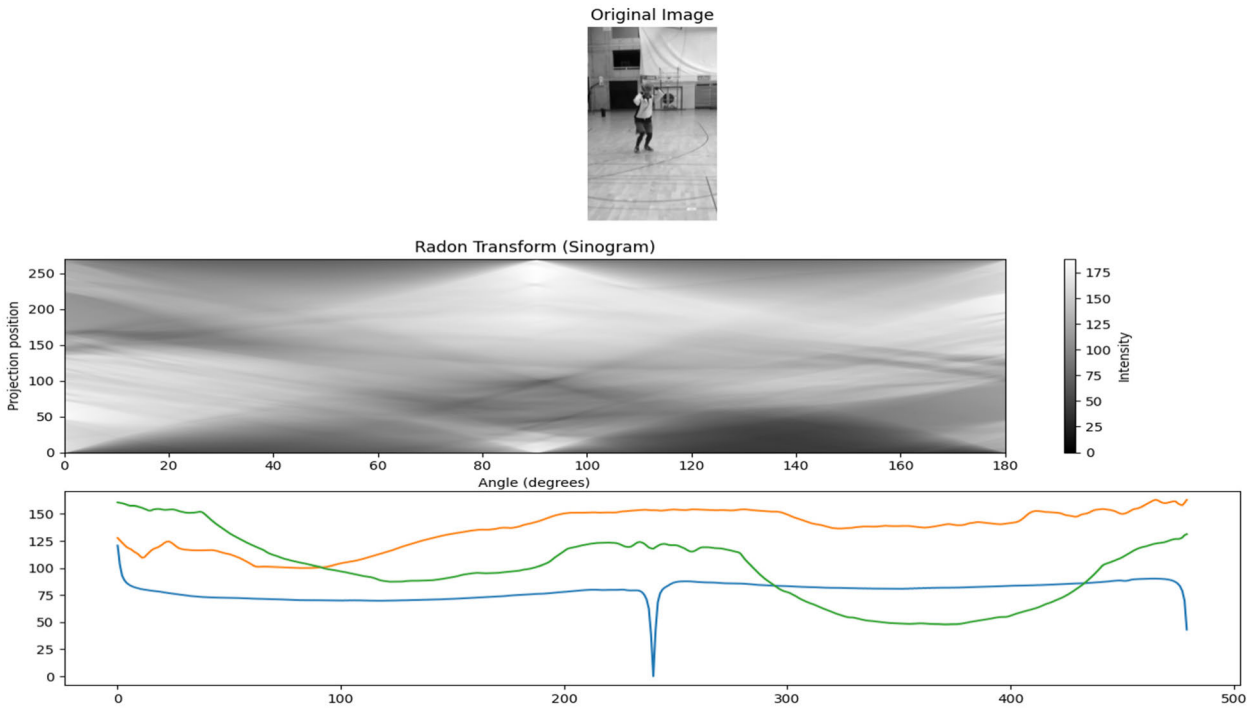

FIGURE 6. Visualization of the Radon transform (sinogram) of the original image, highlighting intensity variations across projection angles and positions.

4) DETECTION OF DISTINCT MOVABLE BODY PARTS FEATURES

In moveable body parts features, only relocated body parts of the human were considered [33]. To identify these body parts features, we considered the moveable section of the human body parts in the preceding frame as the main spot, to crop the given frame path p of size IxJ in the present frame and to estimate the output value as:

$$\hat{S} = ft^{-1}(A \odot ft(n^{\hat{xp}})) \tag{12}$$

where ft^{-1} denotes the inverse Fourier transform, $\odot$ denotes the matrix Hadamard product, n is a correlation, x' is the output value of the marked shape in a certain image and $\hat{S}$ shows the similarity among the candidate portion of the frame and the preceding region. Thus, the present location of the moveable body parts can be identified by obtaining the higher values of $\hat{S}$ as:

$$C = \max(\hat{S}) \tag{13}$$

However, when the transformed regions were recognized, we increased the bonding region across the moving body points, found the pixel's location, and traced additional moveable body parts in the series of frames as:

$$Nb = \sum_{0}^{Mk} NI(Mk) \tag{14}$$

where Nb is the moving body parts vector, Mk is the integer index and NI denotes the location of pixel values. Figure 7. describes the results of moveable body parts features.

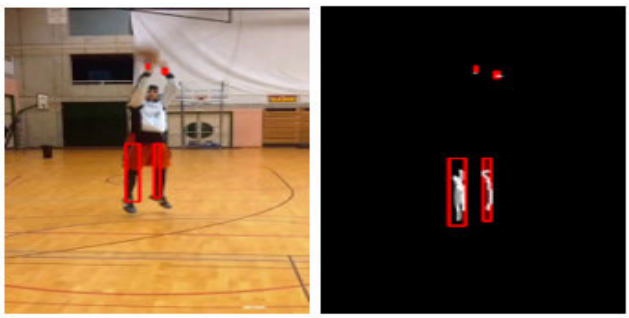

FIGURE 7. Result of the distinct moveable body parts across the basketball player image.

D. KEYPOINTS FEATURE

Keypoint feature extraction is a pivotal module in human pose estimation, targeting 17 anatomical keypoints to represent critical body joints and movements. By employing techniques like the degree of freedom, human landmark detection, and angular point analysis, this module captures both static and dynamic attributes of human motion [34]. These techniques allow for precise identification of joint positions, flexibility in movement, and spatial orientation, enhancing the system's ability to model complex human activities [35]. The integration of these advanced approaches ensures robust keypoint localization, even in challenging scenarios with occlusions or dynamic lighting, providing a solid foundation for accurate posture analysis and event recognition.

1) DEGREE OF FREEDOM (DOF)

To address the Degree of Freedom (DOF) in the context of keypoint features, the technique is fundamental in quantifying human motion by extracting precise angular movements across the body's 17 anatomical keypoints. Each keypoint represents a significant joint or landmark (e.g., head, neck, elbows, knees, etc.), allowing the system to capture three-dimensional angular variations in the x, y, and z-axes [36]. The DOF analysis ensures robust tracking of human posture, accommodating diverse motion patterns. It is particularly crucial in complex scenarios where motion must be measured relative to both local joint movements and global positional changes [37]. By modeling DOF, our method bridges the gap between raw keypoint detection and higher-order feature computation, enhancing the precision of subsequent pose-based classification tasks.

$$D_{of} = \sum_{i=1}^{N} \left[\sqrt{\Delta\theta_{x_i}^2 + \Delta\theta_{y_i}^2 + \Delta\theta_{z_i}^2} . w_i \right] \tag{15}$$

where D_{of} represents the overall degree of freedom feature metric, $\Delta\theta_x$, $\Delta\theta_y$, $\Delta\theta_z$ shows the Changes in angular positions for keypoint i in the x, y, and z-axes, w_i represents the weight factor associated with the importance of mobility of each keypoint and N is the total number of key points analyzed.

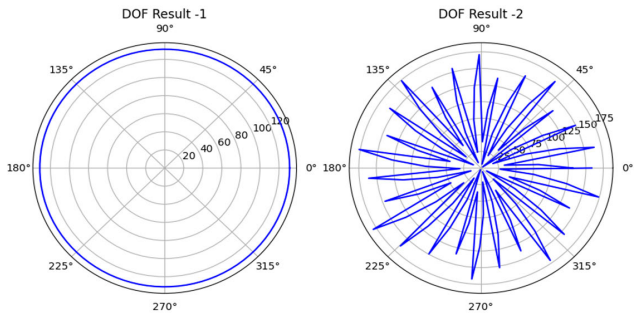

FIGURE 8. Polar representations of Degree of Freedom (DOF) analysis showcasing radial patterns in two scenarios: uniform distribution (left) and directional peaks (right).

2) HUMAN LANDMARK DETECTION

The Human Landmark Detection process plays a crucial role in accurately identifying and extracting significant keypoints on the human body, such as joints and landmarks, for pose estimation and motion analysis [38]. This technique leverages the Hue, Saturation, and Value (HSV) color model to distinguish between various regions of the human body, enabling the effective detection of key points. The HSV model, with its ability to separate chromatic content (hue) from intensity information (value), is particularly well-suited for human landmark detection in environments with varying lighting conditions or complex backgrounds [39].

The HSV algorithm segments the human body into distinct regions by analyzing the hue for identifying skin tones or clothing, the saturation for intensity variations, and the value for brightness. By processing this information, the algorithm maps critical anatomical points, forming a robust foundation for subsequent motion analysis and feature extraction tasks. This method's adaptability and precision ensure the effective identification of key points in diverse scenarios, such as dynamic activities or occluded settings.

$$H = tan^{-1}\left(\frac{\sqrt{3}(G-B)}{2R-G-B}\right), S = \frac{max(R,G,B) - min(R,G,B)}{\max(R,G,B)}, \quad V = \max(R,G,B) \tag{16}$$

where $H = arctan\frac{(B-G)}{(R-G)}$ computes the hue (H), where R, G, and B are the red, green, and blue components of the image pixel. The saturation (S) and value (V) are derived using the max and min values of the RGB channels, where S and V help to distinguish different colors for keypoint detection in the human body.

$$E_{landmark} = \sum_{i=1}^{N} \left[\alpha \cdot \nabla H_i^2 + \beta \cdot \nabla S_i^2 + \gamma \cdot \nabla V_i^2 \right] \tag{17}$$

where E calculates the energy function for detecting keypoints, ∇H, ∇S, ∇V are the gradients of hue, saturation, and value. The parameters α, β, and γ adjust the contribution of each color channel's gradient, allowing fine-tuning for more accurate human landmark extraction.

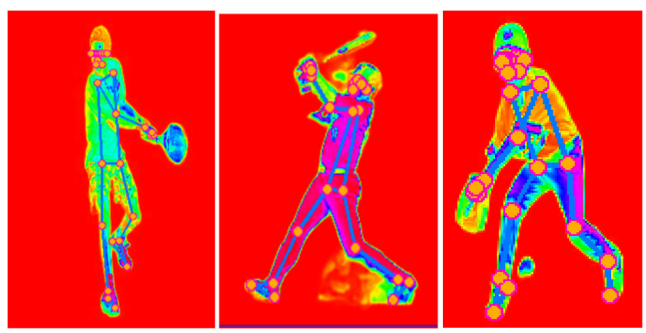

FIGURE 9. Human body landmark detection results using an HSV color map.

3) ANGLES POINTS FEATURES

By examining the angle relationships between keypoints to comprehend posture and movement, the Angles Point Feature is essential to human keypoint recognition. This technique collects exact angle measurements between connected body joints, like elbows, shoulders, and knees, using the MediaPipe algorithm. A reliable foundation for real-time angle computation and keypoint 3D position estimation is offered by MediaPipe [40]. These angular characteristics are essential for applications in activity identification and human pose estimation because they provide information about both static and dynamic body postures.

$$\theta = arccos\left(\frac{\vec{u}.\vec{v}}{\|\vec{u}\| \, \|\vec{v}\|}\right) \tag{18}$$

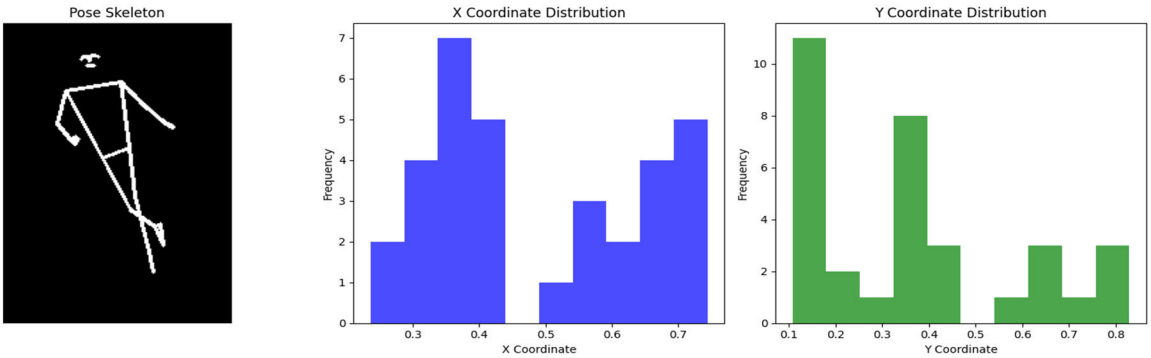

FIGURE 10. Pose skeleton representation with corresponding X and Y coordinate frequency distributions highlighting spatial variability.

where $\vec{u}$ and $\vec{v}$ are vectors formed by adjacent keypoints (e.g., shoulder-to-elbow, elbow-to-wrist). $\|\vec{u}\|$ and $\|\vec{v}\|$ are magnitudes of the vectors. θ is the angle, providing a quantitative measure of joint orientation.

$$E_{angle} = \sum_{i=1}^{N} \left[\alpha_i.(\theta_i - \theta_{ref,i})^2\right] \tag{19}$$

where E_{angle} is the energy function quantifying deviations from reference angles $\theta_{ref,i}$. θ_i represents the detected angle for the i-th joint. α_i shows the weight assigned to the i-th joint based on its significance and N shows the total number of joint angles considered.

E. FEATURE OPTIMIZATION VIA RAY OPTIMIZER

Feature optimization plays a critical role in enhancing the efficiency and effectiveness of any machine learning or computer vision model. In this research, Ray Optimizer is employed to optimize extracted features, offering unparalleled advantages in computational speed and scalability. Its distributed nature allows simultaneous exploration of large and high-dimensional feature spaces, which is particularly important when dealing with complex datasets that include redundant or noisy features [41]. By identifying the most informative subset of features, the Ray Optimizer significantly reduces dimensionality without compromising performance, resulting in faster model training and inference.

The necessity of a feature optimizer stems from the fact that raw feature sets often contain irrelevant or redundant information, which can lead to overfitting, increased computational complexity, and poor generalization. Ray Optimizer systematically refines these features through parallelized gradient-based optimization, ensuring that only the most salient features contribute to the final model. This not only improves predictive accuracy but also makes the model robust and interpretable, enabling it to adapt to diverse real-world scenarios. The optimizer's capability to handle massive datasets and its seamless integration with distributed frameworks make it indispensable in achieving high-performing, scalable solutions in feature-heavy applications.

$$J(F) = \frac{\sum_{i=1}^{n} (y_i - f(x_i; F))^2}{n} + \lambda.\|F\|_1 \tag{20}$$

where $J(F)$ is the objective function to minimize. y_i shows the ground truth for the i-th sample. $f(x_i; F)$ represents the predicted value using features F. λ is the regularization parameter controlling the sparsity. $\|F\|_1$ is the L1-norm of the feature weights to enforce sparsity.

$$\nabla J(F) = \frac{\partial}{\partial F}\left(\frac{1}{n}\sum_{i=1}^{n} (y_i - f(x_i; F))^2\right) \tag{21}$$

where $\nabla J(F)$ represents the gradient of the objective function concerning the feature set F. Computed in a distributed manner across all processing nodes for scalability and efficiency.

F. EVENT CLASSIFICATION VIA NEURO FUZZY CLASSIFIER

Event classification is the final and critical module in our workflow, ensuring that the extracted and optimized features are effectively mapped to their corresponding events. For this purpose, we utilized the Neuro-Fuzzy Classifier (NFC), an advanced hybrid approach that integrates the learning capabilities of neural networks with the rule-based reasoning of fuzzy logic systems [42]. This classifier is particularly advantageous due to its ability to handle uncertainty, nonlinearity, and ambiguity in the feature space, which are inherent challenges in real-world datasets. The Neuro-Fuzzy Classifier dynamically constructs fuzzy rules from input data and iteratively refines them through neural learning algorithms, enabling robust and adaptive classification [43]. By leveraging both human-like reasoning and machine learning efficiency, NFC ensures higher accuracy and reliability even in noisy or incomplete datasets. This makes it indispensable for event classification in applications such as human activity recognition, video analysis, and behavioral studies [44]. Moreover, its scalability allows seamless integration with large datasets, ensuring that the model adapts well to evolving patterns and scenarios.

$$y = \frac{\sum_{k=1}^{K} w_k.B^k}{\sum_{k=1}^{K} w_k} \tag{22}$$

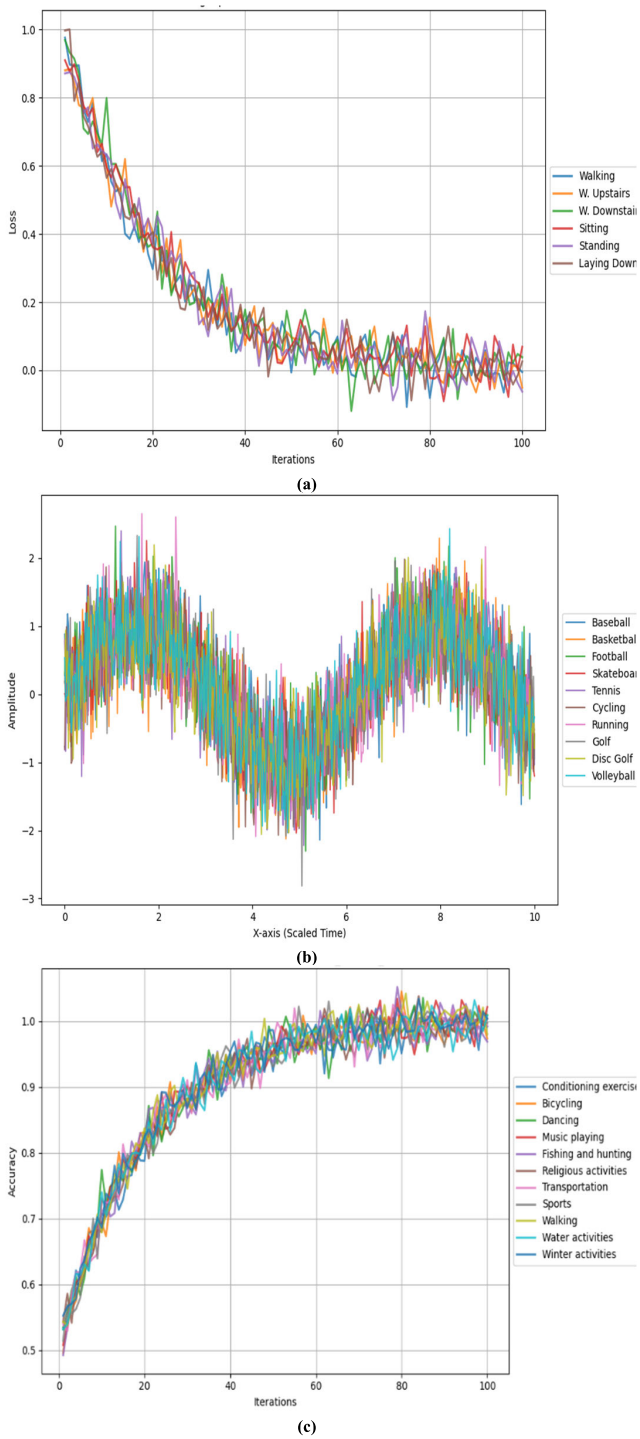

FIGURE 11. Graphical representation of optimized features: (a) Optimized features across the HAR dataset, (b) Optimized features of the UCF Sports Action dataset, (c) Optimized features over the MPII Human Pose dataset.

where $w_k = \prod_{i=1}^{n} \mu_{A_i^k(x_i)}$ is the rule weight based on membership function μ, B^k is the defuzzified output of the k-th rule and K is the total number of fuzzy rules.

$$\Delta\mu_{A_i^k} = -\eta.\frac{\partial E}{\partial \mu A_i^k} \tag{23}$$

where η is the learning rate, E represents the error function defined as $E = \frac{1}{2}\sum(y_{pred} - y_{true})^2$, $and\, \mu A_i^k$ represents the membership function parameter being updated.

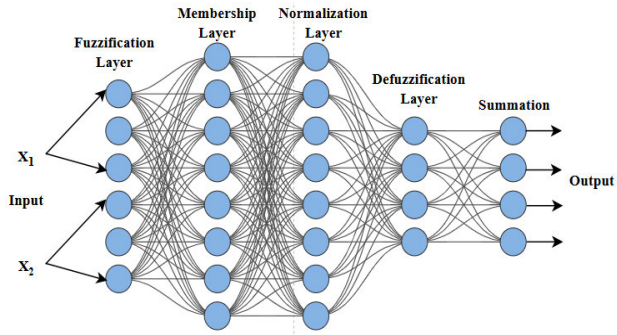

FIGURE 12. Workflow diagram of the neuro fuzzy classifier.

IV. EXPERIMENTAL ANALYSIS AND RESULTS

The leave-one-subject-out (LOSO) cross-validation strategy was employed to evaluate the proposed methodology's performance on three publicly available datasets: UCF Sports Action, HAR, and MPII Human Pose. LOSO is an advanced cross-validation method that uses data from one subject per fold, enabling a comprehensive assessment of the model's ability to generalize effectively across different individuals.

A. DATASET DESCRIPTION

1) UCF SPORTS ACTION DATASET

The UCF Sports Actions dataset [45] consists of about 150 video clips across 10 human action classes, including running, diving, and kicking, performed in sports contexts. It provides annotated sequences with bounding boxes for actor detection, captured under real-world conditions with varied lighting and cluttered backgrounds. The videos are converted into frame images for processing. Its realistic settings and variety make it an essential resource for evaluating action recognition algorithms.

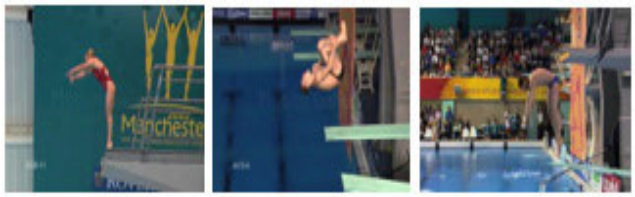

FIGURE 13. Sample images from UCF Sports Action Dataset.

2) HUMAN ACTION RECOGNITION (HAR) DATASET

The Human Action Recognition (HAR) dataset [46] is a comprehensive collection of over 10,000 high-resolution images categorized into 12 action classes, such as running, walking, and jumping. It features diverse, real-world scenarios with complex backgrounds and multi-object scenes, making it ideal for action recognition tasks. Annotated with bounding

Algorithm 1 Proposed Approach

Input: $Iv = \{Video_data1, Video_data2, \ldots\ldots\}$
img_{frames}
Output: $E = \{event_1, event_2, event_3, \ldots, event_N\}$
Method:
Step 1: Preprocessing
$I_f \leftarrow VideoToFrames\ (I_v)$, where $I_f = \{F_1, F_2, \ldots, F_N\}$
$F_{blurred} \leftarrow FloorRemover(F)\ \forall F \in I_f$
$F_{gray} = Grayscale(F_{blurred})$
$S \leftarrow SilhouetteExtraction(F_{gray})$
Step 2: Human Detection
Define the mask for the region of interest:
$M \leftarrow Grabcut(S)$
If *Human presence is detected:*$M \neq \emptyset$ *continues:*
Else *Skip to the next frame.*
Step 3: Full Body Features Extraction
SwitchF_{body}:
Case*1-D Fourier Transform:*
$F_{FFT} \leftarrow FFT(M)$.
Case*2D Energy Heat Mapping:*
$H \leftarrow DenseOpticalFlow(M) + MOG2BackgroundSubtraction(M)$
Case *0–180° Intensity Mapping*:
$F_{Radon} \leftarrow RadonTransform(M)$
Case *Distinct Movable Body Parts*:
$F_{moveable} \leftarrow MovablePartsDetection(M)$
Append all features:
$F_{body} \leftarrow [F_{FFT}, H, F_{Radon}, F_{moveable}]$
Step 4: Keypoints Features
For each$F$ *in*F_{body}:
$DOF \leftarrow DegreeOfFreedom(F)$
$K_{HSV} \leftarrow HSVKeyPoints(F)$
$A_{angles} \leftarrow MediaPipeAngles(F)$
Aggregate keypoints:
$K \leftarrow [DOF, K_{HSV}, A_{angles}]$
Step 5: Feature Optimization
$F_{opt} \leftarrow RayOptimizer(F_{body} + K)$
Where $F_{opt} = \{optimal\ feature\ set\ for\ each\ frame\}$
Step 6: Event Classification
For each F_{opt}:
$E \leftarrow NeuroFuzzyClassifier(F_{opt})$
Append classified events:$E_{detected} \leftarrow E$
Return $E_{detected}$

FIGURE 14. Sample images from HAR Dataset.

boxes and pose information, the dataset supports applications like object detection, pose estimation and behavior classification. Its variety and detailed labeling make it highly suitable for developing robust deep learning-based action recognition systems.

3) MPII HUMAN POSE DATASET

The MPII Human Pose dataset [47] is a comprehensive benchmark for human pose estimation. It consists of 25,000 images, primarily in RGB format, annotated with 40,000 individuals performing a wide range of activities. The dataset includes 16 body keypoints per person and covers diverse poses and camera perspectives. Additionally, it supports various classes of human activities, such as sports, daily tasks, and more. The dataset features complex interactions, making it suitable for evaluating pose estimation and human activity recognition algorithms. It includes multi-person scenarios, ensuring its utility for advancing multi-person pose estimation research.

FIGURE 15. Sample images from MPII Human Pose Dataset.

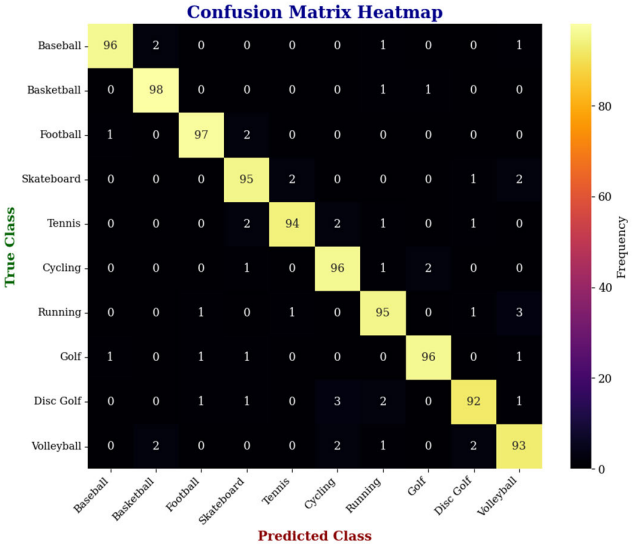

FIGURE 16. Confusion matrix of the UCF Sports Action Dataset.

B. HARDWARE PLATFORM

Experiments were conducted on datasets using an Intel Core i5-12500H CPU, 24GB RAM, and an RTX-3050 GPU (4GB RAM) to evaluate real-time performance. Both CPU and GPU resources were utilized under varying computational demands to assess the efficiency of keypoint detection and posture estimation. Profiling and benchmarking provided insights into processing speed, resource utilization, and scalability. These analyses highlighted the proposed methods' robustness, optimization potential, and suitability for practical applications in dynamic and resource-constrained environments.

C. EXPERIMENTAL SETUP AND EVALUATION

The study's analysis of the proposed system's performance across the UCF Sports Actions, HAR, and MPII Human

TABLE 1. Precision, Recall, and F1-score are evaluated using the UCF Sports action dataset.

Classes	Precision	Recall	F1-Score
Baseball	0.9600	0.9600	0.9600
Basketball	0.9600	0.9800	0.9700
Football	0.9700	0.9700	0.9700
Skateboard	0.9500	0.9500	0.9500
Tennis	0.9400	0.9400	0.9400
Cycling	0.9600	0.9600	0.9600
Running	0.9500	0.9500	0.9500
Golf	0.9600	0.9600	0.9600
Disc Golf	0.9400	0.9200	0.9300
Volleyball	0.9300	0.9300	0.9300
Mean	**0.9530**	**0.9520**	**0.9525**

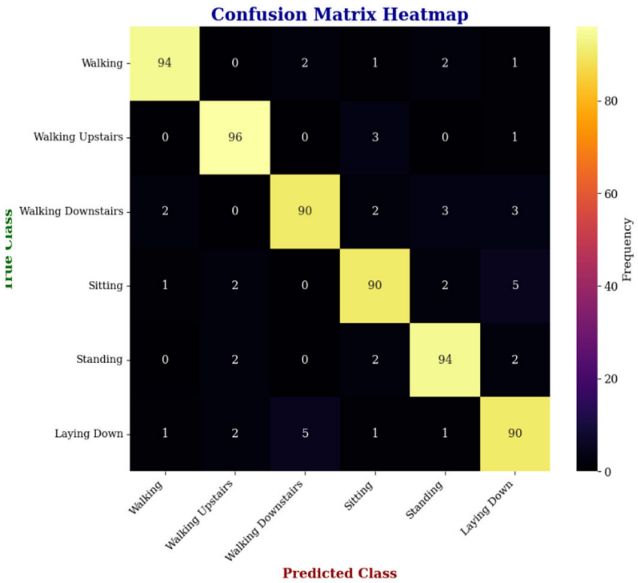

FIGURE 17. Confusion matrix of the HAR Dataset.

TABLE 2. Precision, Recall, and F1- score are evaluated using the HAR dataset.

Classes	Precision	Recall	F1-Score
Walking	0.9592	0.9400	0.9495
Walking Upstairs	0.9412	0.9600	0.9505
Walking Downstairs	0.9278	0.9000	0.9137
Sitting	0.9091	0.9000	0.9045
Standing	0.9216	0.9400	0.9307
Laying Down	0.8824	0.9000	0.8911
Mean	**0.9235**	**0.9233**	**0.9233**

Pose datasets demonstrated its effectiveness. For the UCF Sports Actions dataset, confusion matrices are presented in

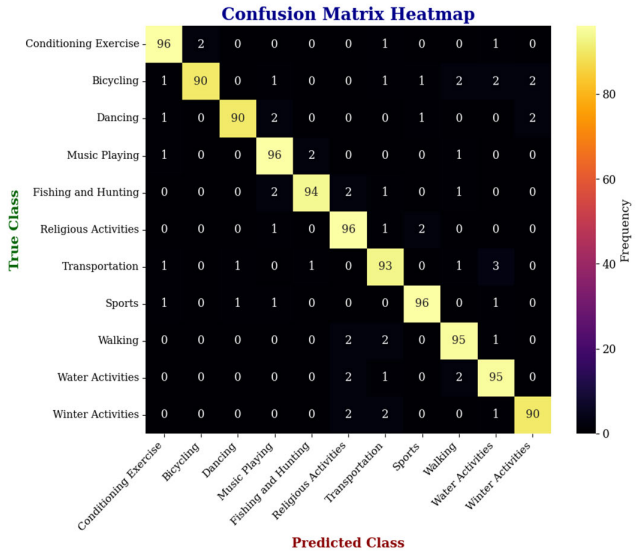

FIGURE 18. Confusion matrix of the MPII Human Pose Dataset.

TABLE 3. Precision, Recall, and F1-score are evaluated using the MPII Human Pose dataset.

Classes	Precision	Recall	F1-Score
Conditioning Exercise	0.9505	0.9600	0.9552
Bicycling	0.9783	0.9246	0.9375
Dancing	0.9783	0.9375	0.9574
Music Playing	0.9320	0.9600	0.9458
Fishing&Hunting	0.9691	0.9400	0.9543
Religious Activity	0.9231	0.9600	0.9412
Transportation	0.9118	0.9300	0.9208
Sports	0.9600	0.9600	0.9600
Walking	0.9314	0.9500	0.9406
Water Activities	0.9135	0.9500	0.9314
Winter Activities	0.9575	0.9474	0.9524
Mean	**0.9459**	**0.9450**	**0.9451**

Figure 16, yielding an accuracy value of 94.82%, with the corresponding Precision, Recall, and F1-scores detailed in Table 1.

We calculate Precision, Recall, and F1-score (F-measure) using the Neuro-fuzzy classification algorithm, applying the relevant formulas. These metrics are then visualized through curves that depict their respective values.

$$Precision = \frac{TP}{TP + FP} \tag{24}$$

$$Recall = \frac{TP}{TP + FN} \tag{25}$$

$$F1 - Score = \frac{2\,(Precision \times Recall)}{(Precision + Recall)} \tag{26}$$

TABLE 4. Human body key point detection accuracy.

Body Keypoints	Distance	UCF Sports Action (%)	Distance	HAR (%)	Distance	MPII Human Pose (%)
FH	10.2	88	9.1	86	10.0	87
LE	9.9	89	8.8	87	9.6	89
RE	10.1	88	9.0	86	9.8	88
NS	9.5	92	8.7	90	9.3	91
NC	9.8	91	8.9	90	9.5	92
LS	11.3	85	10.2	83	10.8	84
RS	10.9	87	9.8	85	10.4	86
LE	12.1	84	11.0	82	11.7	83
RE	11.8	86	10.5	84	11.3	85
LW	8.5	80	7.9	78	8.9	81
RW	9.3	83	8.2	80	9.0	82
CH	13.0	92	11.8	90	12.2	91
SP	12.7	90	11.5	89	11.9	89
LHJ	10.9	88	9.7	85	10.5	87
RHJ	10.8	87	9.6	84	10.4	86
LA	10.6	82	9.8	80	10.3	81
RA	10.5	81	9.7	79	10.2	80

(*FH=Forehead, LE=Left Ear, RE=Right Ear, NS=Nose, NC=Neck, LS=Left Shoulder, RS=Right Shoulder, LE=Left Elbow, RE=Right Elbow, LW= Left Wrist, RW=Right Wrist, CH=Chest, SP=Spine, LHJ=Left Hip Joint, RHJ=Right Hip Joint, LA=Left ankle, RA=Right ankle*).

TABLE 5. Time complexity and execution time of the algorithms.

Methods	Time Complexity	Execution Time (s)	Energy Efficiency (J)	Error Margins (%)
Pre-processing	O($n^{3/2}$), n=pixels	0.310	5.4	Low (3.1%)
Human Detection	O(m *$\sqrt{n}$), m=pixels, n=frames	0.660	7.1	Moderate (5.6%)
Full Body Features	O (n^3), n=keypoints	0.870	8.3	Low (4.2%)
Keypoints Feature	O (n log^2 n), n=keypoints	0,770	9.0	Moderate (6.0%)
Feature Optimization	O(log n*m), m=iterations	0.950	6.8	Minimal (3.9%)
Event Classification	O(m*n^2), m=features, n=samples	0.960	7.5	Moderate (5.3%)

For the HAR dataset, the confusion matrix results are provided in Figure 17, showing an accuracy of 92.33 %, with the Precision, Recall, and F1-scores in Table 2.

The MPII Human Pose dataset's performance is summarized in the confusion matrix presented in Figure 18, achieving an accuracy of 93.63%, and the related Precision, Recall, and F1 scores are shown in Table 3. Additionally, Table 10 offers a comparative analysis of our system with other widely used models, highlighting the robustness and reliability of our approach.

We thoroughly compared the accuracy, precision, recall, and F1-score based on examining our suggested technique using the UCF Sports Action, HAR, and MPII Human Pose datasets. The accuracy comparisons across datasets are shown in Figure 19. With the greatest values found in the UCF Sports Action dataset, we specifically find that the accuracy, precision, recall, and F1-score show a continuous increase throughout the datasets. This graphical display clearly shows the performance of the Neuro Fuzzy classification algorithm, confirming our method's scalability and resilience across many kinds of data.

D. EXPERIMENT: BODY PART DETECTION ACCURACIES

To evaluate the effectiveness and precision of identifying human body components, a revised approach is introduced, incorporating normalized distances and a decision criterion.

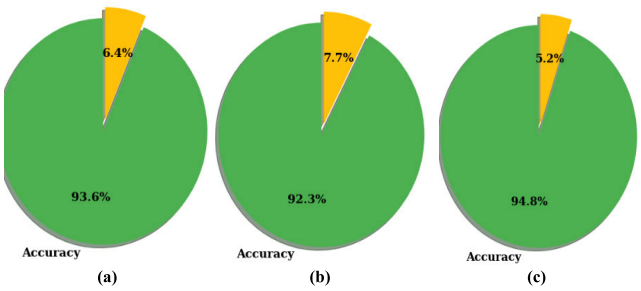

FIGURE 19. Accuracy comparison across all datasets.

The distance calculation between detected keypoints and ground truth values is now expressed as:

$$R_i = \sqrt{\frac{1}{N}\sum_{m=1}^{N}\left(\frac{I_m}{S_m} - \frac{G_m}{S_m}\right)^2} \tag{27}$$

where G_m represents the ground truth keypoint, I_m is the detected keypoint, S_m normalizes the spatial scale, and N is the total number of components evaluated. This formulation ensures robustness across varying dataset resolutions and body part sizes. To quantify the percentage of correctly identified body components, the following accuracy metric is

used:

$$RA = \frac{100}{n} \sum_{n=1}^{N} \begin{cases} 1 & if\ R \leq T \\ 0 & otherwise \end{cases} \tag{28}$$

Table 4 displays the distances from the ground truth dataset in columns 2, 4, and 6, and the human component identification accuracies over the UCF Sports Action, HAR, and MPII Human Pose datasets in columns 3, 5, and 7, respectively. Table 5 summarizes the time complexity and execution times of the algorithms used.

TABLE 6. Comparison of the proposed method with SOTA over datasets.

Methods	UCF Sports Action	HAR	MPII Human Pose
Wang et al [48]	91.2%	--	--
Raptis et al [49]	88.3%	--	--
Lan et al [50]	89.9%	--	--
Chen et al [51]	--	89.7%	--
Yana et al [52]	--	90.3%	--
Luo et al [53]	--	91.4%	--
Jain et al [54]	--	--	86.5%
Ionescu et al [55]	--	--	87.8%
Sapp et al [56]	--	--	89.2%
Proposed Method	**94.82%**	**92.33%**	**93.63**

Table 6 presents a comparison between the proposed method and several leading algorithms across three datasets: UCF Sports Action, HAR, and MPII Human Pose. The results demonstrate that our method outperforms competing approaches in all cases. Specifically, the UCF Sports Action dataset achieves an impressive 94.82% accuracy. For the HAR dataset, it reaches 92.3%, and for the MPII Human Pose dataset, it attains 93.63%. These results highlight the proposed system's superior performance and generalizability across multiple datasets.

V. DISCUSSION/LIMITATIONS

While the proposed system achieves high accuracy, its performance may degrade during real-time processing due to the computational intensity of keypoint detection and event classification, especially in resource-constrained environments. The system may struggle with accurate event classification when the subject's pose deviates significantly from the training data, particularly in extreme or unusual angles, which can impact its robustness in dynamic, real-world applications.

VI. CONCLUSION AND FUTURE WORK

This paper presents a robust framework for HPE and SEC, achieving superior accuracy in real-world scenarios. By integrating advanced feature extraction techniques, such as the Radon Transform, with machine learning models like Neuro-Fuzzy classification, the system demonstrates enhanced performance over existing approaches. The proposed method shows significant potential for applications in dynamic environments, where accurate human motion detection and event classification are crucial. The system achieves classification accuracies of 94.82% on the UCF Sports Actions dataset, 92.33% on the HAR dataset, and 93.63% on the MPII Human Pose dataset. Future work could focus on improving the system's real-time processing capabilities and enhancing robustness in highly cluttered or occluded environments. Additionally, expanding the dataset diversity and exploring multi-modal inputs could further improve the model's generalizability.

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